

Java Concepts and Explanations

Variables and Data Types

Variables store data values, while data types define the kind of data that can be stored, such as int, float, char, boolean, and String.

Operators

Operators perform arithmetic, logical, relational, and assignment operations on variables and values.

Conditional Statements

Conditional statements like if, else if, switch, and else control program flow based on specified conditions.

Loops

Loops such as for, while, and do-while execute a block of code repeatedly until a condition is met.

Arrays

Arrays are data structures used to store multiple values of the same type in a single variable.

Methods

Methods are reusable blocks of code that perform specific tasks and improve code organization.

Object-Oriented Programming

Java follows OOP principles including Encapsulation, Inheritance, Polymorphism, and Abstraction to create scalable applications.

Classes and Objects

A class is a blueprint for creating objects, while an object is an instance containing data and behavior.

Exception Handling

Exception handling uses try, catch, finally, throw, and throws to manage runtime errors effectively.

Collections Framework

The Collections Framework provides classes like ArrayList, LinkedList, HashSet, and HashMap for efficient data storage and manipulation.

Multithreading

Multithreading enables concurrent execution of multiple threads, improving application performance and responsiveness.

JDBC

JDBC (Java Database Connectivity) allows Java applications to connect and interact with databases.

File Handling

Java provides classes for reading, writing, and managing files using streams and file APIs.